

Mastercam Assignment 1

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Course: CAPT1638

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Mastercam Assignment 1

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1. Introduction

The Mastercam Assignment 1-Mill involves the machining a specific part using the Tormach 440 Mill using Mastercam software in creating toolpaths, generating NC programs, and ultimately machining a physical part to document observations, errors, corrections, and conclusions from the machining process.

2. Wireframe Geometry

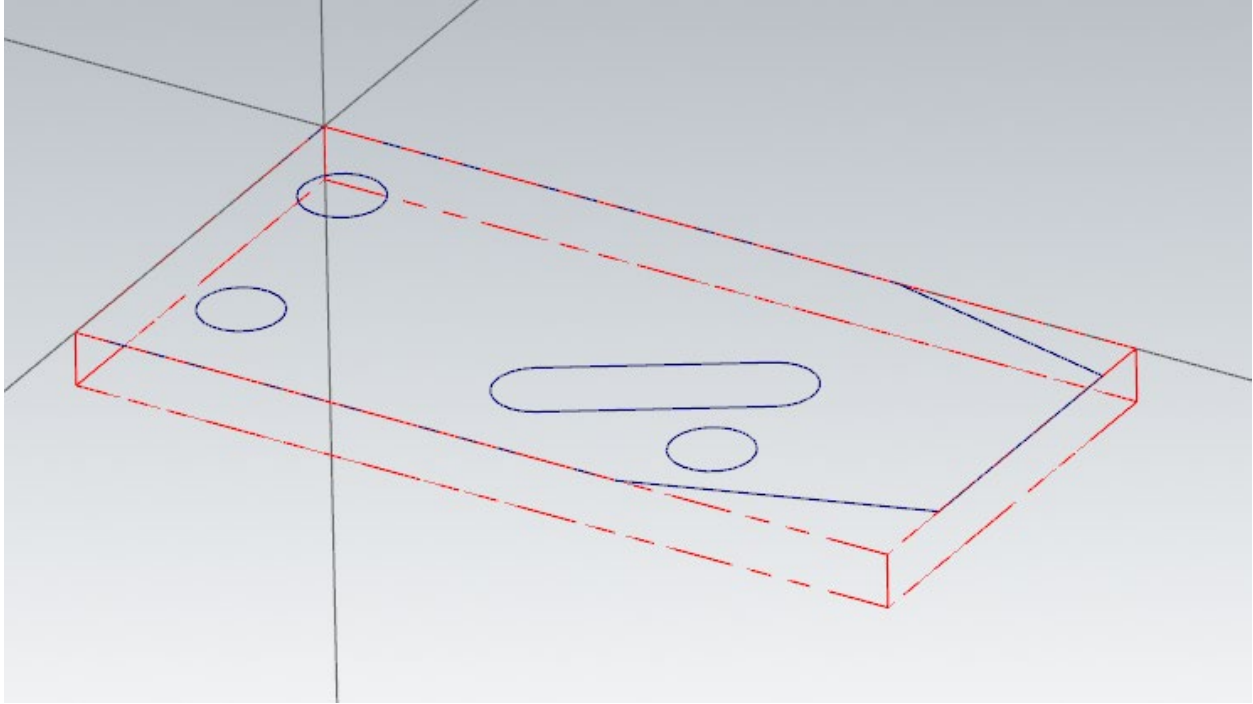


Figure 1 Wireframe Geometry

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3. Stock Setting

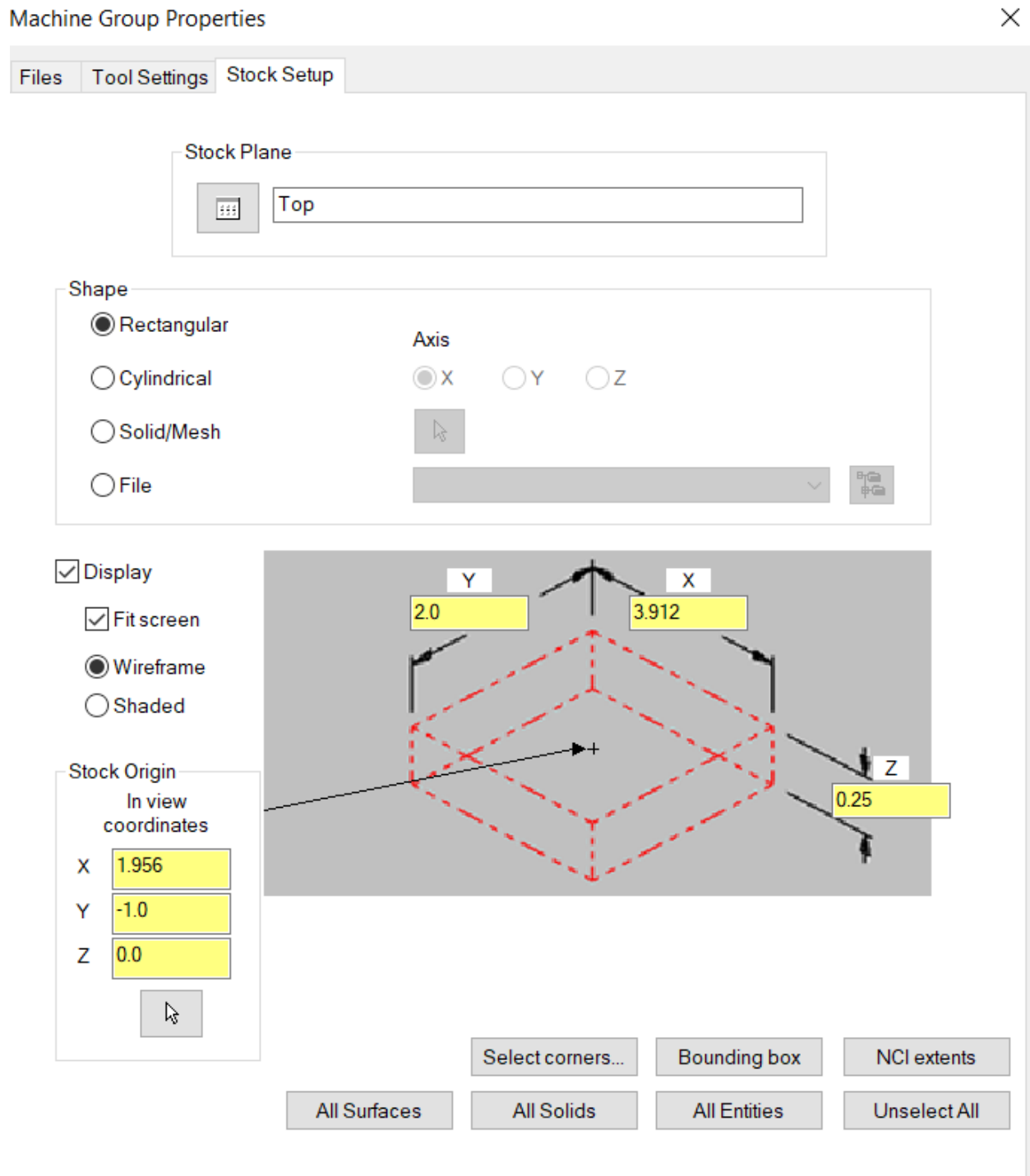


Figure 2 Stock Setting

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4. Backplots of all operations

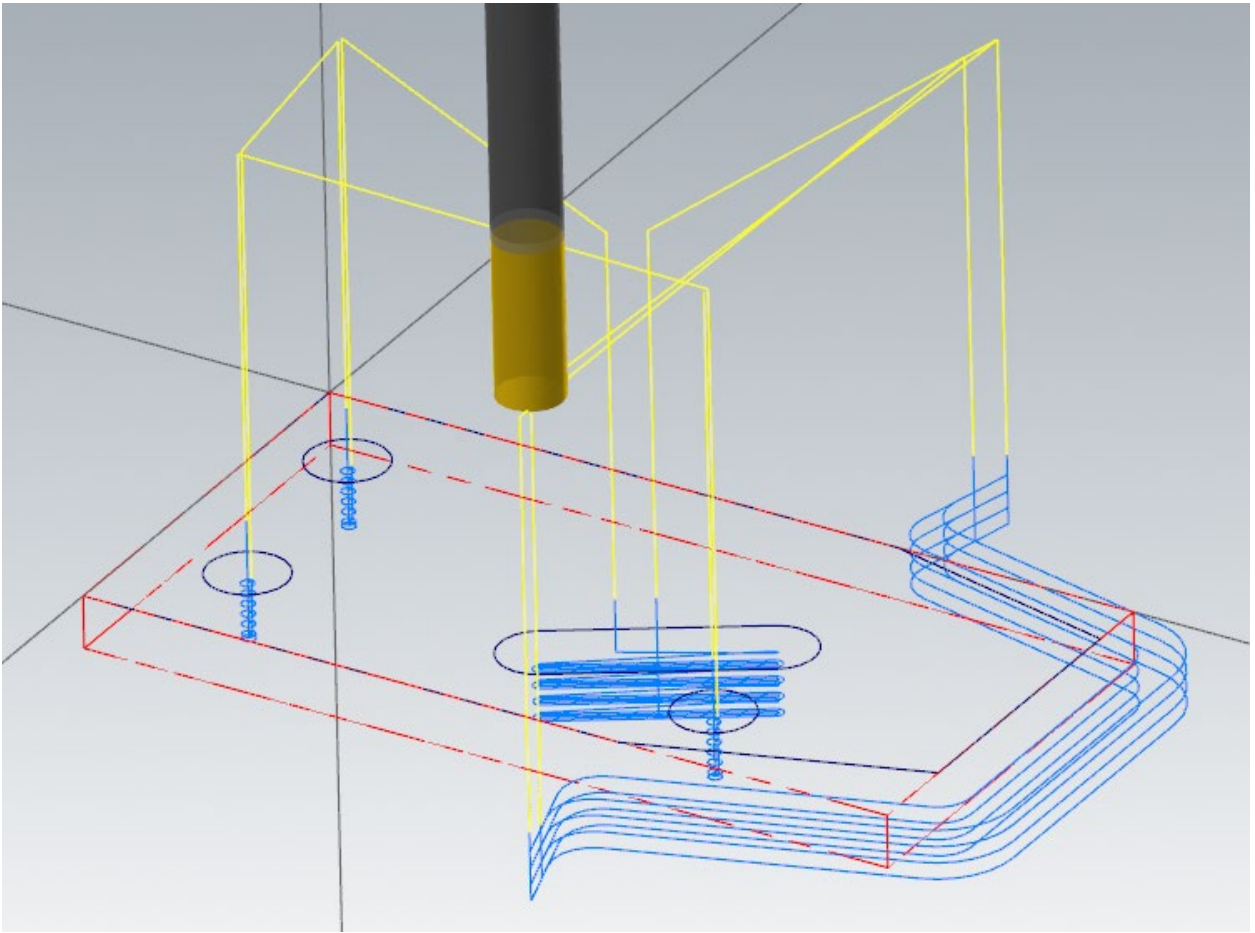


Figure 3 Backplots of all operations

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5. Verification of all operations

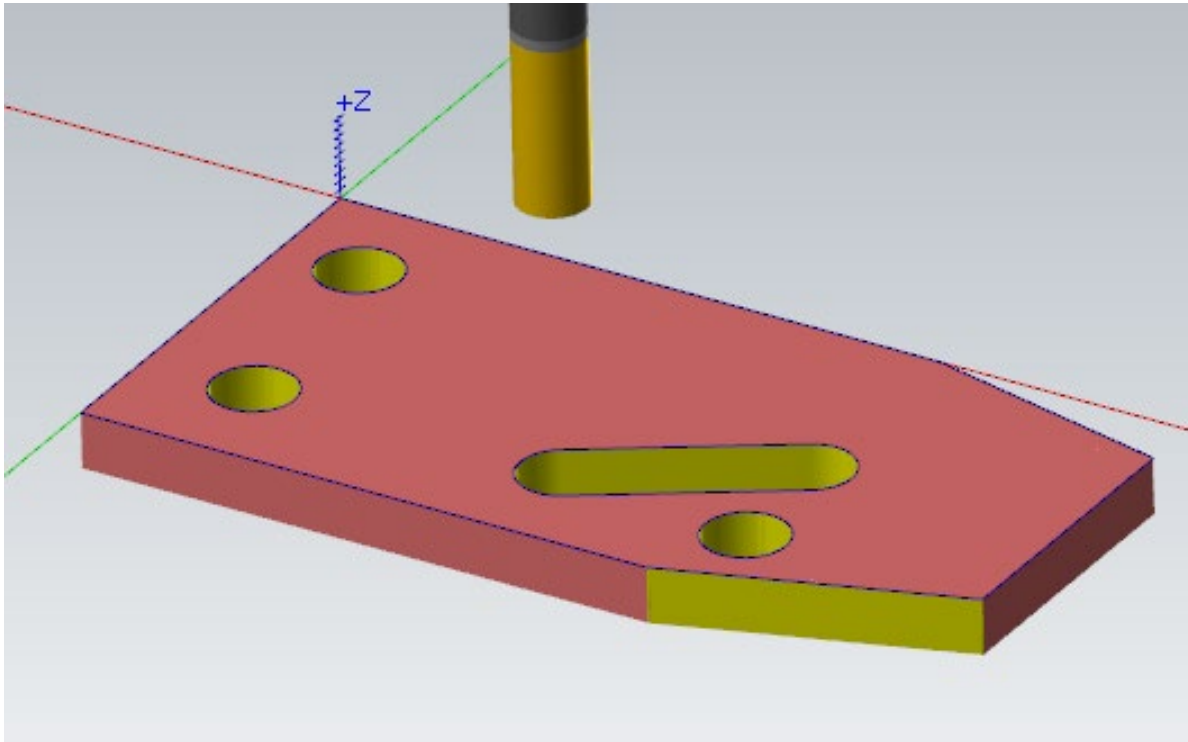


Figure 4 Verification of all operations

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6. Dimensional Check

6.1 Labeled Part Drawing and Dimensional Check Sheet

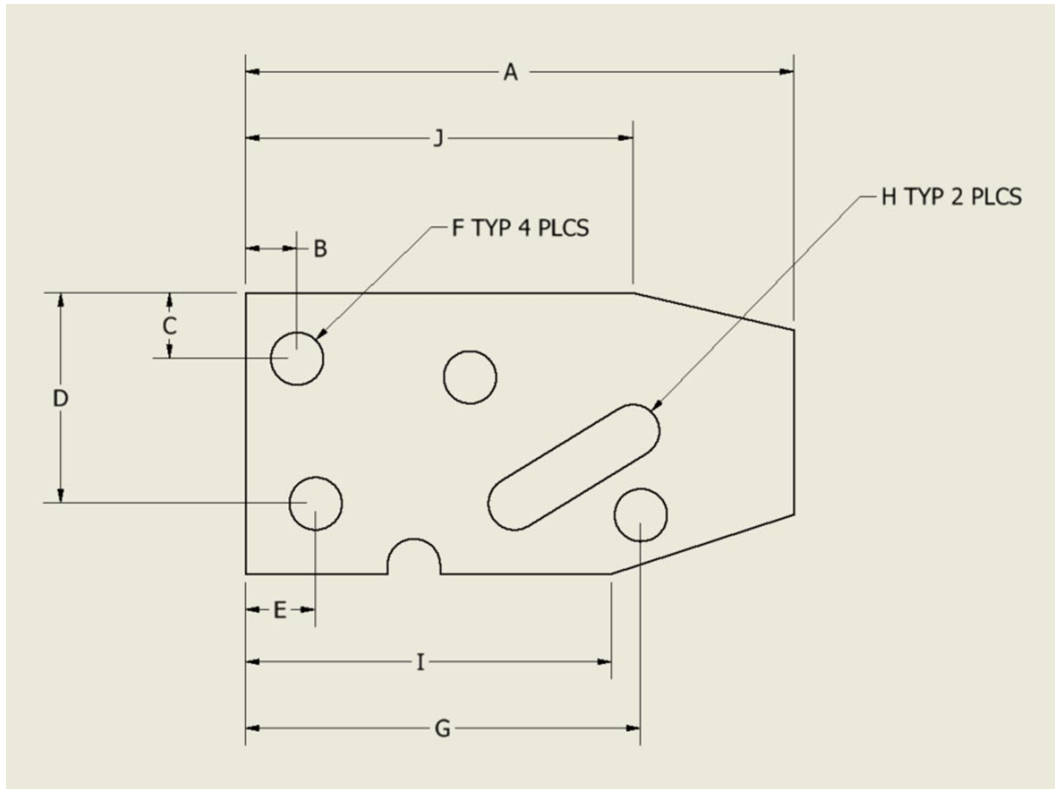


Figure 5 Labeled Part Drawing

6.2. Dimensional Check Sheet

Dimensional Check Sheet							
Part Name:	Metrology Plate	Job No:	CAPT1638	Date:	Mar/29/2024		
Part No:	M1	Rev:	00	Units:	US-Customary		
Label	Specified Dimension	Specified Tolerance	Measured Value		Tool Used	OK	Reject
A	3.912	+/- 0.005	3.9230				
B	0.368	+/- 0.005	0.3678				
C	0.468	+/- 0.005	0.4938				
D	1.500	+/- 0.005	1.5253				
E	0.500	+/- 0.005	0.5093				
F	Ø0.375	+/- 0.005	Ø0.3705				
G	2.818	+/- 0.005	2.8308				
H	Ø0.376	+/- 0.005	Ø0.3630				
Inspector Name:		Ahmed Ibrahim Ibrahim Adem Emmanuel Muhiga			Section:		
Notes:							

Figure 6 Dimensional Check Sheet

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7. Images of the Part Being Cut



Figure 7 Images of the Part Being Cut



Figure 8 Images of the Part Being Cut

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Figure 9 picture of finished part

8. Observations/Errata/Conclusions

8.1 Observations

While cutting the holes the chips were long narrower chips but when machining the corners the chip were very short chips

8.2 Errata

During the circle mill operation, the end mill jammed mid operation as observed in the video due to high forces being applied on the tool, as advised by the lab instructor. A solution was to replace the circle mill operation with a Helix Bore operation using a helical motion as it cuts down in the hole; this applies less force on the tool since there's no plunging involved.

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8.3 Conclusions

The aim of this Mastercam assignment was to machine a specific part using the Tormach 440 Mill using Mastercam software in creating toolpaths, generating NC programs, and ultimately machining a physical cut the part and did a dimensional check. By the end of the project, we had a cut part that was within tolerance.